

Problem 10.21

An investment is made for one year in a fund with

$$a(t) = a + bt + ct^2.$$

The nominal rate of interest earned during the first half of the year is 5% convertible semi-annually. The effective rate of interest earned for the entire year is 7%. Find $\delta_{0.5}$.

Solution

Since an accumulation function satisfies $a(0) = 1$,

$$a = 1.$$

Thus,

$$a(t) = 1 + bt + ct^2.$$

A nominal annual rate of 5% convertible semi-annually corresponds to a half-year effective rate of

$$\frac{0.05}{2} = 0.025.$$

Therefore,

$$a(0.5) = 1.025.$$

Substituting $t = 0.5$ gives

$$\begin{aligned} 1 + 0.5b + 0.25c &= 1.025, \\ 2b + c &= 0.10. \end{aligned}$$

The effective annual interest rate is 7%, so

$$a(1) = 1.07.$$

Hence,

$$\begin{aligned} 1 + b + c &= 1.07, \\ b + c &= 0.07. \end{aligned}$$

Solving the two equations yields

$$b = 0.03, \quad c = 0.04.$$

Therefore,

$$a(t) = 1 + 0.03t + 0.04t^2$$

and

$$a'(t) = 0.03 + 0.08t.$$

The force of interest at time t is

$$\delta_t = \frac{a'(t)}{a(t)}.$$

At $t = 0.5$,

$$\begin{aligned}\delta_{0.5} &= \frac{0.03 + 0.08(0.5)}{1 + 0.03(0.5) + 0.04(0.5)^2} \\ &= \frac{0.07}{1.025} \\ &\approx 0.06829268293.\end{aligned}$$

Thus,

$$\boxed{\delta_{0.5} \approx 6.8293\%}.$$